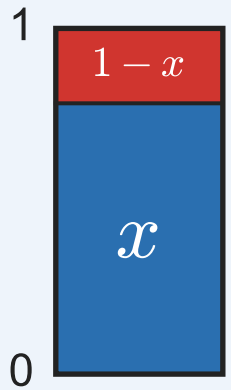


EVOLUTION

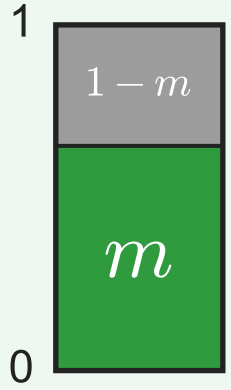
Replicator dynamics

$$\dot{x} = sx(1-x)\left(\pi_L(x,m) - \pi_H(x,m)\right)$$

Payoff matrix

$$\Pi(m) = (1-m)\begin{bmatrix} R_0 & S_0 \\ T_0 & P_0 \end{bmatrix} + m\begin{bmatrix} R_1 & S_1 \\ T_1 & P_1 \end{bmatrix}$$

Exploitation of the environment with harvest effort e_L or e_H



ENVIRONMENT

$$\dot{m} = s\epsilon\left[(\lambda-\mu)\,m(1-m) - qm\left(e_L x + e_H(1-x)\right)\right] + \text{stochasticity}$$